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Software Engineering

COURSE INFORMATION

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2026 Team



Shahid Manzoor



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Essential Information

- 3 credits
- Literature:
 - Sommerville, 10th ed.
 - *Software Requirements*, Wiegers & Beatty, 3rd ed.
- Partial gamified + Roleplaying **(Project Based)** course



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Course Structure

- **Lectures**
 - Theory / class exercises
- **Activities**
 - To help fixing concepts / detect misunderstandings
 - Require active participation
- **Group project** (*participation mandatory + 1 meeting/week*)
 - To experience a “real” scenario
 - You must use your free time
 - Group & Individual deliverables



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Let me hear your experience

Why (Software) Projects Fail?



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Course Goals

- SE is about dealing with large-scale projects (agile is not)

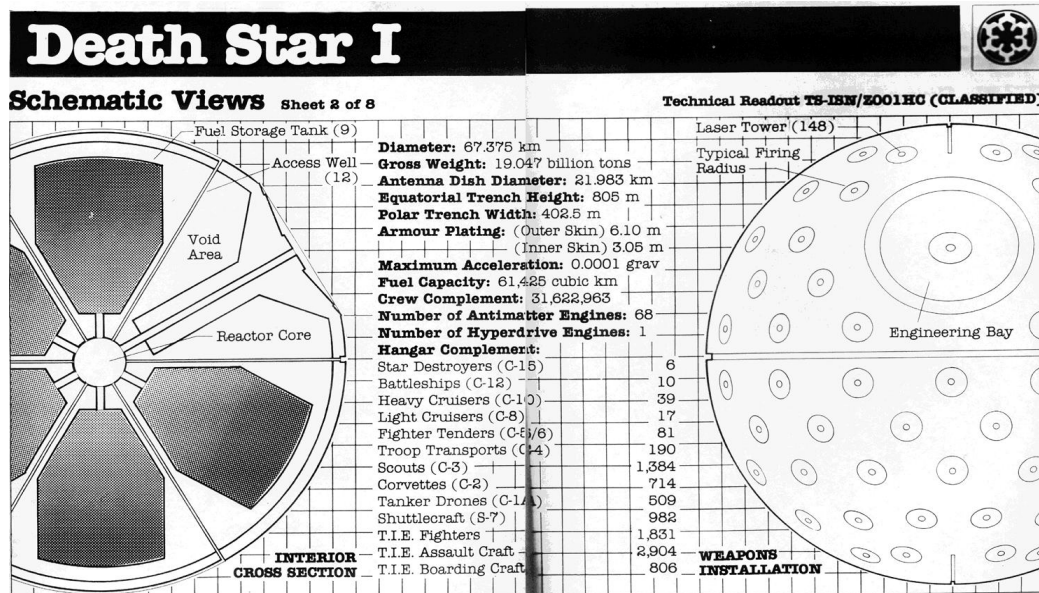




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Course Goals

- We cover aspects related with **processes** such as:
 - Requirements definition and specification
 - Prototypes and models
 - Formal specifications

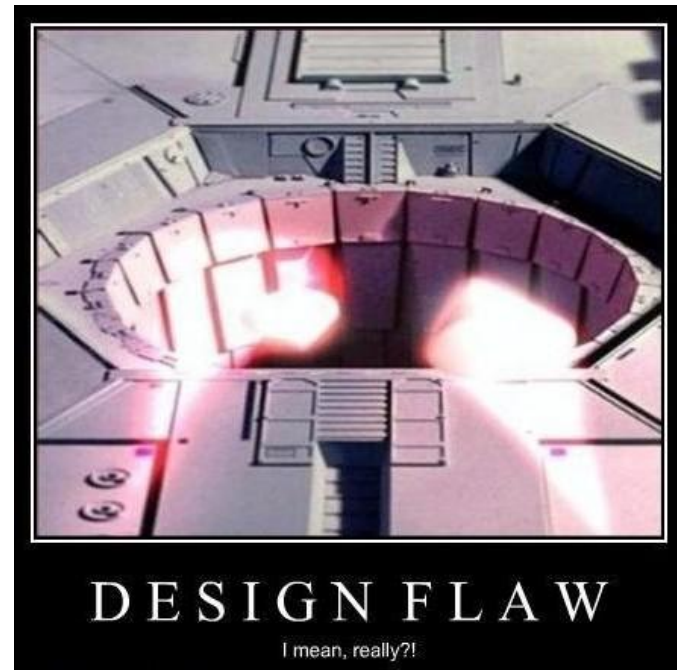




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Course Goals

- We cover aspects related with **quality**:
 - Design methods
 - Maintenance
 - Validation and verification
 - Quality assurance





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Course Goals

- We cover **the human side** of software development:
 - Project planning
 - Project management
 - Human resources





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Course Goals

- SE is about people working together in engineering

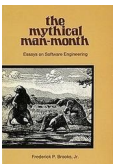
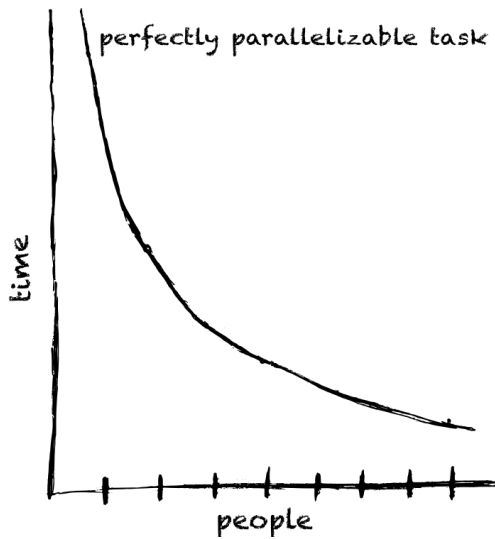




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Course Goals

- SE is about people working together in engineering
- Engineering is complex



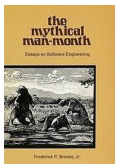
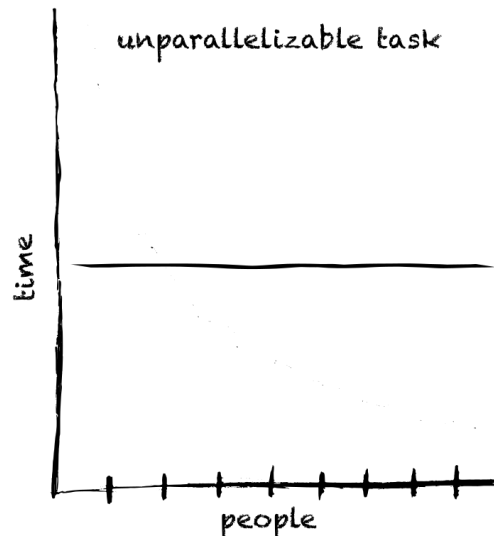
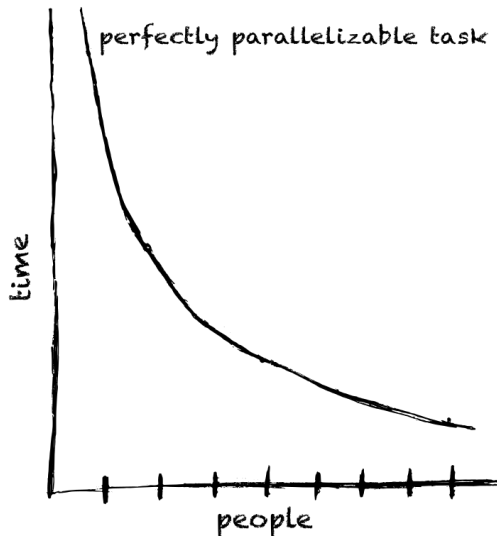
The Mythical Man-Month: Essays on Software Engineering *by Fred Brooks*



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Course Goals

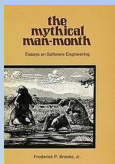
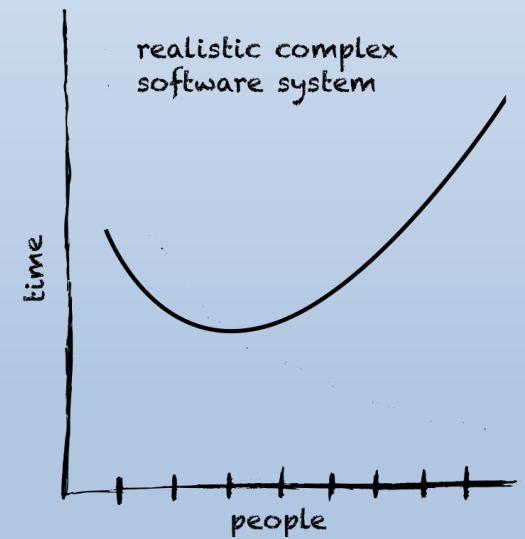
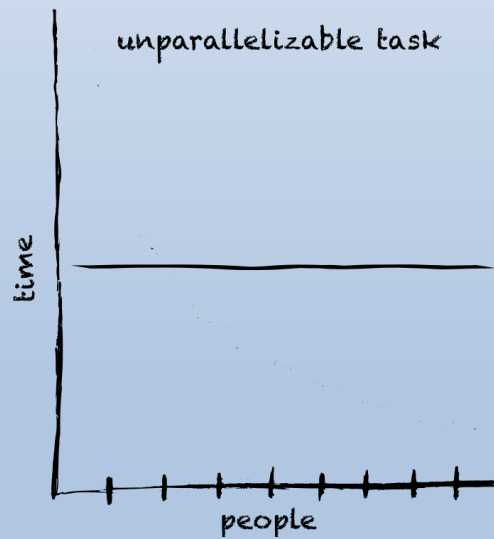
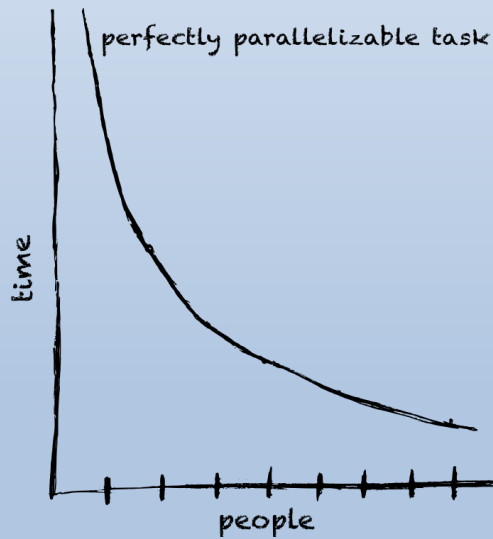
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Course Goals

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The Mythical Man-Month: Essays on Software Engineering *by Fred Brooks*



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Other SE Courses

Requirements in
Agile Development

Software Engineering and Project Management

Advanced
Software Design

Software
Testing

Maintenance
Programming

Complex IT Systems in
Large Organisations

Software Engineering Project



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Course Structure



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Grading (exam) – 2hp

- Course book (specified chapters)
 - Lectures
 - Activities
-
- The exam has forty (40) points and you need 21
 - Three (3) hours duration.
 - Grade: pass (*Godkänd* - G) / fail (*Underkänd* - U)



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Our collaboration



Teaching and learning at Uppsala University

Visions for educational activity and development





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Our collaboration

Good communication is a key aspect of (any) the course

- – Please, you are more than welcome to contact me! Anything related!

Email communication

- – I will do my best to answer your emails promptly (put Jennifer Goss in CC)
- Better if you write through Studium – *it generates an email for you*
- Please, use a relevant subject line - *[SEPM] Topic*
- Be complete. If you need help with anything, indicate at least 3 things you tried before writing – *it saves us time and helps me to have a better idea about how to help you*

Passing by my office

- – We can easily communicate via email, or if an in-person meeting is preferred, I would be happy to arrange it. Please feel free to reach out, and I will set aside specific times and days for us to meet in person.



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Valuable Attributes

- Academic integrity Commitment to
 - learning Cooperation & collaboration
 - Personal initiative
 - Take responsibility
 - Punctuality Reflective thinking
 - Strong work ethic
-
- **Have fun!**



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Questions

?



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The Impossible Project

- You will build a small 2-players game, like...

▼ Medelsvårt

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X tur

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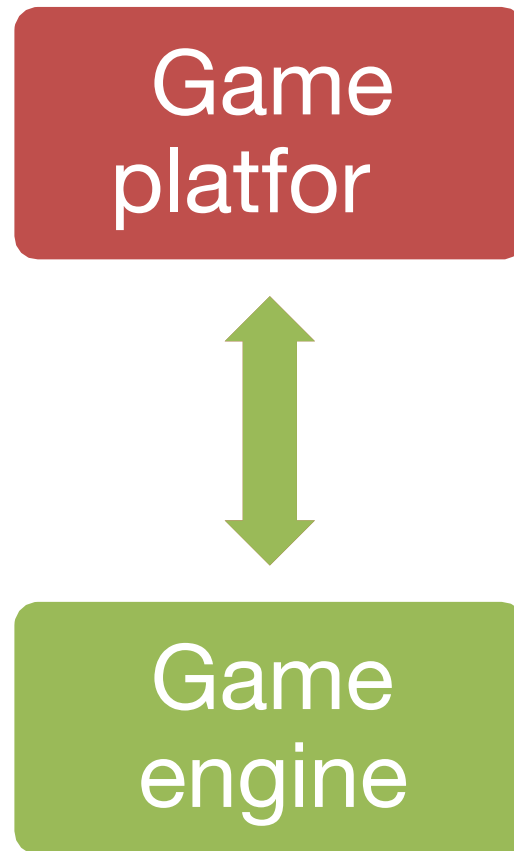
STARTA OM SPELET



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- Main components

The Impossible Project





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The Impossible Project

- Agile, using Scrum methodology + Kanban boards
- Scrum Coach
 - Retrospective / team evaluation
 - 1 Meeting (1h) *usually* on Fridays every 2 weeks
- Project Owner / Client
 - Initial meeting / Discuss progress



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The Impossible Project

- **Phase 1**
 - Initial design and development of 1 component
- **Phase 2**
 - Completing the project
 - Proposal:
 - Outsourcing of the remaining component
 - Verification and integration



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The Impossible Project



**More details in Project Session 1: 2024-
09-05**



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Key points!

- **Attendance volunteer except:**
 - Project participation
 - End Sprint meetings with SC
 - Evaluation (written exam + project progress)
- **To read carefully (in Studium):**

⋮	▼ Course information (course guidelines, project details, gamification, schedule)	✓	+	⋮
⋮	📄 Course details	✓		⋮
⋮	📄 Course schedule	✓		⋮
⋮	📄 Initial questionnaire Sep 1	✓		⋮
⋮	📄 Project Details	✓		⋮